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Deep Learning-Based Postprocessing to Enhance Subseasonal Soil Moisture Forecasts Over Europe

Noelia Otero[1], **Atahan Özer**[1], **and Jackie Ma**[1]

[1]Applied Machine Learning Group, Fraunhofer Heinrich-Hertz Institute, Berlin, Germany

Key Points:

- We present a hybrid modeling framework for postprocessing subseasonal forecasts of soil moisture
- We use Monte Carlo dropout to generate probabilistic forecasts, providing uncertainty estimates alongside improved predictions
- The deep learning models outperform the baselines, enhancing the forecast skill at longer time scales

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

N. Otero,
noelia.otero.felipe@hhi.fraunhofer.de

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Abstract Accurate forecasts on subseasonal (S2S) timescales are essential for the preparation and mitigation of the impacts of high-impact events, such as flash droughts. To improve the accuracy of soil moisture forecasts—a critical factor in identifying flash droughts—we present a hybrid modeling framework that combines dynamical forecasts from the European Centre for Medium-Range Weather Forecasts with deep learning (DL) models. This approach not only corrects biases in numerical weather prediction models but also improves spatial resolution, increasing the accuracy of S2S forecasts. By using deterministic inputs, such as the ensemble mean and spread, we further assess the uncertainty of forecasts through dropout neural networks via Monte Carlo sampling. Our results demonstrate that the DL models outperform baseline methods, offering skillful S2S forecasts of soil moisture. This advanced hybrid framework provides more accurate soil moisture predictions, ultimately supporting improved strategies for managing and mitigating the impacts of flash droughts.

Plain Language Summary Accurate subseasonal soil moisture forecasts are essential for improving predictions of flash droughts, which, unlike traditional droughts, develop and intensify over shorter timescales. In this study, we introduce a hybrid modeling approach that combines numerical weather predictions with deep learning (DL) models. This approach not only corrects errors in the subseasonal forecasts but also enhances their spatial resolution, providing more detailed and reliable predictions. Our results demonstrate the potential of DL models to significantly improve the accuracy and usefulness of subseasonal forecasts of soil moisture, which can aid in better preparedness and mitigation strategies for flash droughts.

1. Introduction

Subseasonal to seasonal (S2S) forecasting is crucial for planning in sectors such as energy, water management, and agriculture (White et al., 2017). However, S2S forecasts are challenging due to limited predictability between short-range weather forecasts (~10 days) and seasonal climate forecasts. The limitations in forecast accuracy arise from both the inherent unpredictability of the chaotic Earth system and errors in numerical models, which are subject to multiple sources of uncertainty, including uncertainty in initial conditions and model parameterizations (Vitart & Robertson, 2018; Vitart et al., 2017). These uncertainties induce systematic and random errors that rapidly amplify over time (Rasp & Lerch, 2018).

Postprocessing relies on traditional statistical techniques like bias correction (Vannitsem et al., 2021), but recent advances in machine learning (ML) and deep learning (DL) have shown significant improvements (Rasp & Lerch, 2018; Vannitsem et al., 2021). ML stands out for its capacity to generalize spatially and temporally, to include extensive predictor variable sets, and to integrate physical insights into models (McGovern et al., 2017). Previous works have shown a superior performance of ML for postprocessing operational short- to medium-range ensemble forecasts over classical methods (Bouallègue et al., 2024; Finn, 2021; Grönquist et al., 2021; Höhlein et al., 2024; Rasp & Lerch, 2018), but its application to S2S postprocessing remains limited (Horat & Lerch, 2024; Scheuerer et al., 2020).

Sectors like water resource management, agriculture, and emergency response rely heavily on drought forecasting. Soil moisture is a widely used indicator for drought identification due to its essential function in regulating the exchange of water and heat within the land-atmosphere feedback process (AghaKouchak, 2014). Given its critical role in controlling the dryness ecosystem (Seneviratne et al., 2010), accurate predictions of soil moisture would lead to better drought monitoring and forecasting systems, in particular for flash droughts which, unlike traditional droughts, are characterized by their rapid onset and intensification (Otkin et al., 2018). Moreover, skillful soil moisture forecasts at the subseasonal scale can also benefit a range of applications

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including agricultural planning as shown in Togneri et al. (2022), wildfire prediction and risk assessment (Dirmeyer et al., 2018), and improved atmospheric forecasts due to enhanced land-atmosphere coupling (Dirmeyer et al., 2019). These impacts further highlight the importance of improving S2S soil moisture forecasts.

Motivated by the success of ML in postprocessing weather forecasts and the increasing demand for more effective techniques to enhance the accuracy of drought predictions, we demonstrate the capability of DL for post-processing S2S forecasts of soil moisture. Recent studies show that DL in hybrid models enhances the S2S soil moisture forecast skill while retaining information about soil moisture persistence and temporal autocorrelation learned from observational data, particularly at shorter lead times (Ma & Yuan, 2023; Xu et al., 2023). Our approach extends previous work by using DL not only to correct errors but also to downscale low-resolution S2S forecasts into higher-resolution predictions, addressing a key limitation of current S2S models. Similar efforts to improve soil moisture S2S forecasting with hybrid DL architectures have also been made recently by Xu et al. (2023), who demonstrated the benefits of hybrid modeling to improve both the surface and root zone levels S2S forecasts over global and local areas. This work seeks to provide further insights into the potential of DL techniques for S2S postprocessing.

The remainder of this paper is structured as follows: Section 2 introduces the data set and describes the methods used in the study. Section 3 presents illustrative examples of the results from the post-processing framework, followed by the conclusions in the final Section 4.

2. Data

For this study, we selected the ECMWF S2S model from the S2S multimodel database (Vitart et al., 2017), which produces reforecasts along with its operational real-time forecasts, running twice a week on Mondays and Thursdays. These reforecasts consist of 11 ensemble members (10 perturbed members and one control member), providing predictions with a lead time of up to 46 days at a spatial resolution of 1.5° and a daily or 6-hourly output frequency depending on the variable.

The operational forecasts, that is, real-time forecasts, which serve as the test data for this study, consist of 51 ensemble members (50 perturbed members and one control member). Note that recent updates, including an increase in the number of perturbed ensemble members, have been implemented. For further details, we refer the interested reader to the ECMWF model information available at https://confluence.ecmwf.int/display/S2S/ECMWF+Model. We use the reforecast data set spanning from 2001 to 2020 and only the perturbed members of both the forecasts and reforecasts. To assess the model performance, forecasts from 2021 are used, incorporating ECMWF model cycles Cy47r1, Cy47r2, and Cy47r3, which were introduced progressively in May and October of 2021.

We use the ERA5 reanalysis (Hersbach et al., 2020) as the reference data set, which has been previously used within the context of flash droughts (Li et al., 2024; Xu et al., 2024; Yuan et al., 2023). The ERA5 data offer soil moisture at four depths (0–7, 7–28, 28–100, and 100–289 cm). For the purposes of this study, only the first two layers (0–7 and 7–28 cm) are used to derive a 20-cm soil moisture profile, aligning with the top 20-cm soil moisture profile (hereinafter, SM20) provided by the S2S data set. While ERA5 may share some of the model biases inherent in S2S forecasts, it remains a reliable reference data set due to its land and atmospheric data assimilation processes, which enhance the accuracy of soil moisture estimates and other environmental variables (Dutra et al., 2021). The target SM20 from ERA5 was obtained at the original spatial resolution of 0.25° and aggregated to daily values, as the low-resolution input from an S2S database. Additional daily features tested in our experiments include mean 2 m temperature (t2m), 10 m wind speed (ws10), and total precipitation (tp).

2.1. Methods

2.1.1. Data Processing

The downscaling task is limited to the European domain (35° −70°N, −10°W-25°E), a region heavily impacted by drought in recent decades (Naumann et al., 2021). We use daily aggregates from the reforecasts to train the model, resulting in 2080 samples (forecast time or initialization dates). Based on preliminary results, we chose deterministic input data, specifically the ECMWF ensemble mean and spread (standard deviation), as using 10 perturbed ensemble members did not improve performance. This aligns with previous studies that also used the

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ensemble mean (Horat & Lerch, 2024; Rasp & Lerch, 2018, e.g.). Limiting inputs to two deterministic variables also reduced computational costs.

The input data are structured as a tensor with dimensions *L, N, C, H,* and *W*, where *L* represents the lead times, *N* refers to the ensemble representation (reduced to the mean and standard deviation), *C* denotes the feature channels (atmospheric variables), and *H* and *W* correspond to the spatial coordinates (latitude and longitude). We applied bilinear interpolation to prescale inputs and used min-max normalization to ensure numerical stability during neural network training.

Reforecasts from 2001 to 2018 were used for model training, with 2019 and 2020 reforecasts reserved for validation, totaling 1,664 and 416 samples. We trained each model with low-resolution inputs, gradually incorporating additional meteorological variables (e.g., t2m, ws10, and tp) to assess their impact on predictive accuracy. More details on the training procedure are available in Supporting Information.

2.1.2. Modeling and Architectures

We formally define the problem as a spatial regression task where we seek a mapping function $\mathbf{f}(\theta, \mathbf{X})$, parameterized by θ, from the input space $\mathcal{X} = \{L, N, C, H, W\}$ to the output space $\mathcal{Y} = \{L, H', W'\}$. The dimensions H' and W' represent the downscaled resolutions. Following Stormer (Nguyen et al., 2023), who focus on predicting weather dynamics, we aim to predict the forecast bias $\delta = \mathcal{Y} - \hat{\mathcal{X}}$, where $\hat{\mathcal{X}}$ represents the ensemble mean of the SM20 reforecasts.

To quantify the model uncertainty, we apply the Monte Carlo (MC) dropout method as a Bayesian approximation following Gal & Ghahramani (2016), allowing us to estimate the epistemic uncertainty by performing stochastic forward passes through the network. While dropout is typically applied during training, it can also be used during inference to obtain stochastic predictions. In this way, probabilistic forecasts are generated by applying MC dropout during the testing phase. As shown in Scher & Messori (2021), dropout ensemble techniques can lead to an improvement in accuracy. Similar perturbation methods, such as MC dropout have been presented in previous works (Clare et al., 2021; Garg et al., 2022). Here, MC dropout is used to create 10 random realizations to match with the 10 ensembles, consistently with the number of ensembles used for training. A schematic illustration of the methodological postprocessing framework is provided in Figure 1.

We tested several architectures, starting with the popular U-Net (Ronneberger et al., 2015) that has been widely used for different weather and climate applications, such as weather forecasting (Weyn et al., 2020), explainability of precipitation extremes (Otero & Horton, 2023), and downscaling (Horat & Lerch, 2024; Xin et al., 2024). For our implementation, we have adopted the U-Net from Nguyen et al. (2023), which incorporates residual blocks to enhance performance on tasks involving spatial-temporal climate data. While this version supports attention mechanisms, we opted not to include the attention layer to minimize computational costs during training. The model follows a typical U-Net structure with an encoder-decoder architecture, where the encoder captures multiscale features through a series of downsampling layers, and the decoder reconstructs the output with upsampling layers. Skip connections between corresponding encoder and decoder layers help retain spatial details. We further modified the architecture to handle the temporal information. Such temporal embedding mechanism incorporates lead time information directly into the network. By encoding temporal dimensions, the model can process time-varying inputs.

Our modeling framework also includes an adaptive Fourier neural operator (AFNO) (Guibas et al., 2021). This architecture showed great potential for weather forecasting (Pathak et al., 2022), and it has been recently tested for climate downscaling tasks (Sinha et al., 2024; Yang et al., 2023). As shown in previous studies, the AFNO architecture leverages the power of Fourier transforms to capture long-range dependencies and spatial-temporal patterns (Pathak et al., 2022). The model employs a deep architecture, consisting of multiple AFNO layers, each equipped with multilayer perceptrons that enhance its capacity to capture complex features. The incorporation of sparsity and hard thresholding techniques enables the model to manage the complexity of spectral features effectively, making it well-suited for a range of applications in image analysis. We adapted the original AFNO implementation in NVIDIA Modulus (https://github.com/NVIDIA/modulus) and as for the U-Net, we further include lead time embeddings, allowing the AFNO to integrate temporal information and enhance its performance in sequential data tasks. As an alternative architecture, we chose SegFormer introduced by Xie et al. (2021) within a semantic segmentation framework for our downscaling task. The model uses a hierarchical

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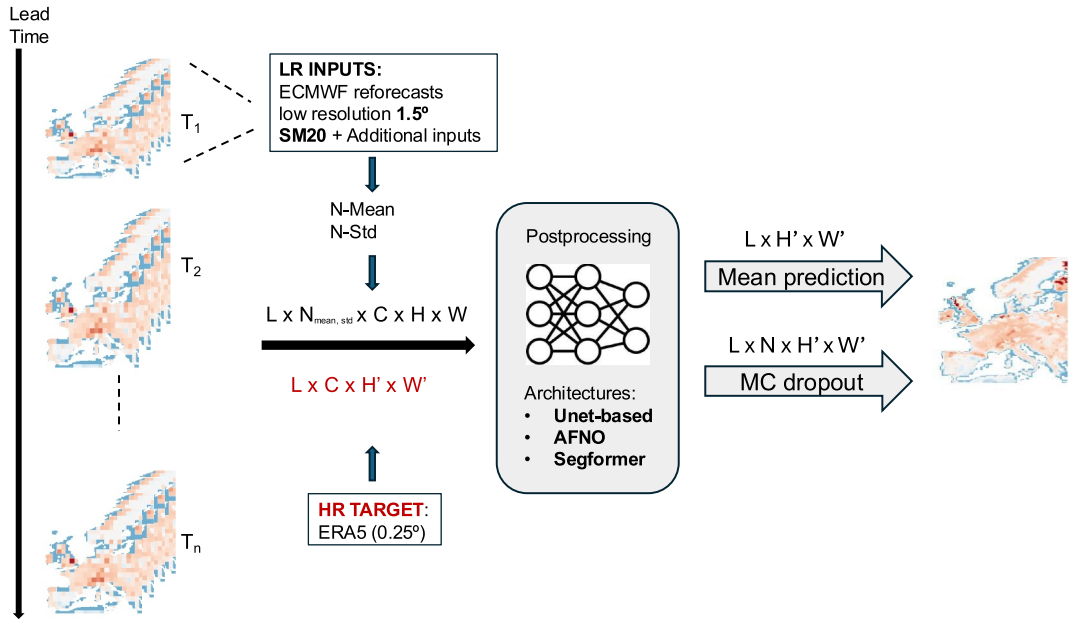


Figure 1. Schematic representation of the downscaling framework: Low-resolution inputs at multiple lead times (L) are transformed into input tensors of shape $L \times N_{\text{mean,std}} \times C \times H \times W$, where C is the number of channels, and H and W are the spatial dimensions. These inputs are processed by a neural network model (adaptive Fourier neural operator, U-Net, or SegFormer) to generate high-resolution (HR) outputs. To estimate predictive uncertainty, Monte Carlo dropout is applied during inference, producing N stochastic samples. This results in output tensors of shape $L \times N \times H' \times W'$, where H' and W' denote the spatial dimensions of the HR output. The final predictions are obtained by computing the mean across the N samples.

encoder to extract multiscale features using convolutional layers and efficient self-attention mechanisms. These layers are designed to handle spatial and temporal patterns effectively. The encoder divides the process into multiple stages, each applying overlapping patch extraction, convolutional embeddings, and attention layers to progressively reduce the spatial dimensions while capturing richer feature representations. Here, time embeddings are optionally integrated, allowing the model to incorporate temporal information from lead times. The original architecture does not include any dropout layers. Thus, we added a single dropout layer to the final multilayer perceptron block to obtain the MC samples.

We performed several ablation experiments by varying the dropout rate and model sizes. Based on preliminary results, we selected a configuration for each model that offered a good balance between computational efficiency and performance. Additional details are provided in Section 1 of the Supplementary Information.

2.1.3. Evaluation

For the evaluation, we use the climatology, as a standard competitive benchmark commonly used in subseasonal to seasonal forecasting (Horat & Lerch, 2024; Mouatadid et al., 2023; Nathaniel et al., 2024). Following a similar procedure presented in (Rasp et al., 2024), a probabilistic climatology of the target is constructed by grouping several years into 10 ensembles, consistent with the reforecasts setup and also aligned with the forecast validity time (Bouallègue et al., 2024). Given the 20-year period (2001–2020), each ensemble is created by grouping 2 years together and averaging over the forecast time dimension.

Additionally, we use the S2S forecasts bilinearly interpolated to the finer target resolution without bias correction as a baseline. Deterministic metrics include bias, root mean square error (RMSE), and mean squared error, which quantify systematic errors and overall accuracy. We also use the anomaly correlation coefficient (ACC) to assess the model's ability to represent observed anomalies. For probabilistic evaluation, we use the continuous ranked probability score (CRPS), which measures the difference between the hindcast and observed cumulative density functions (Hersbach et al., 2020), providing an overall quality of the forecast distribution. The continuous ranked probability skill score (CRPSS) evaluates whether the predictions improve or worsen with respect to the

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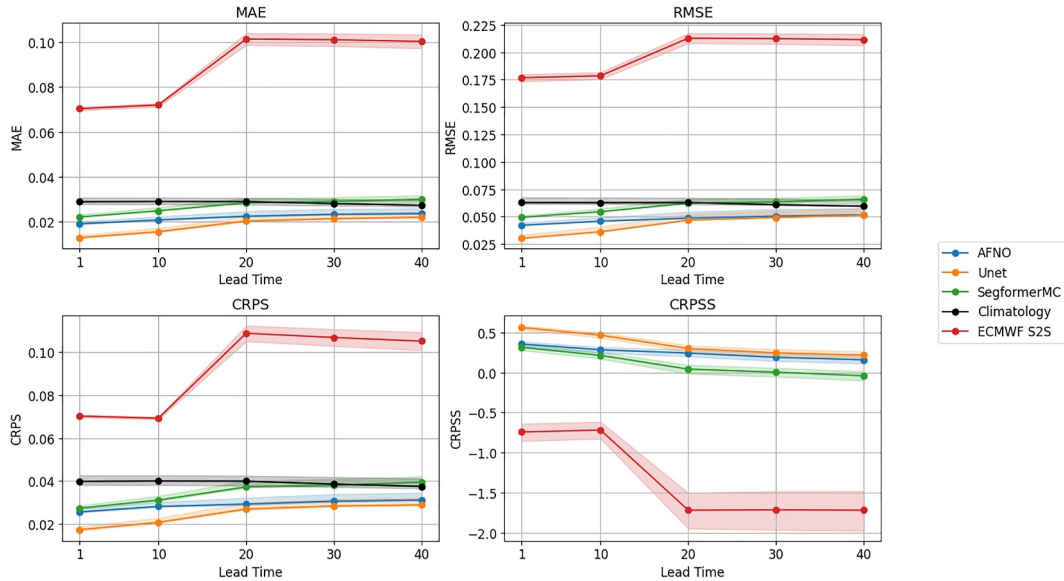


Figure 2. Skill scores spatially averaged over the whole European domain for the following lead times: 1, 10, 20, 30, and 40. Color lines indicates the DL models, the interpolated ECMWF S2S model, and the climatological benchmark. Shaded areas represent the 5% confidence interval over 10,000 bootstrap samples.

climatology. These metrics were computed for both validation and test data sets, respectively, to ensure robustness and generalization. Unless stated otherwise, the results presented correspond to the test data set (i.e., real-time forecasts). The evaluation was conducted using 10 ensemble members, consistent with the training ensemble, but the deterministic input forecasts (mean and standard deviation) are not expected to affect results significantly.

Due to the smaller sample size of the testing data set, the uncertainty was further quantified by resampling the test forecasts 10,000 times using bootstrap sampling (Vitart, 2014). Skill metrics were computed for each resampled set, and the 5%–95% confidence interval was subsequently derived from the distribution of these samples.

3. Results

We trained the models on daily inputs and by aggregating the data into weekly averages to assess the effect of data aggregation on model performance (see Supporting Information S1). The results indicated no significant differences between the two approaches. Consequently, only the daily forecast results are presented here, while additional information on the performance of weekly forecasts can be found in Supporting Information S1.

3.1. Forecasts Skill of Daily Forecasts

Figure 2 displays the model's performance when using only SM20 as an input. The DL models, consistently achieve significantly lower MAE and RMSE values across all lead times than the ECMWF S2S model. This demonstrates the superior accuracy of the DL models, particularly as lead time increases, where the ECMWF S2S model exhibits higher errors (Figure 2, top). Despite that the climatology offers a competitive benchmark, the DL models provide higher deterministic skill scores, even at longer lead times. A similar pattern is found for the probabilistic metrics, such as CRPS and CRPSS, with the DL models outperforming the ECMWF S2S model. The negative CRPSS values for the ECMWF S2S model indicate poorer skill in the reforecasts than the observed climatology. Overall, the DL models achieve CRPSS values greater than 0 across all lead times, indicating higher skill than the baseline, except for SegFormer, which shows a decline to negative values at the longest lead times.

As shown in Figure 2, a temporal inhomogeneity is observed in the ECMWF S2S model between days 10 and 20. This pattern aligns with the temporal bias identified during the analysis of the raw data (Figure S4 in Supporting Information S1). Such temporal inhomogeneity could be attributed to a change in the ECMWF model resolution, which is T639 before day 16 and T319 at longer lead times (Lorenz et al., 2024).

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Overall, Figure 2 indicates the effectiveness of only using SM20 as an input for DL models, resulting in significantly enhanced performance and forecasting skill. Notably, adding more predictors did not yield significant performance gains (Table S1 and Figure S1 in Supporting Information S1). Our experiments suggest that the additional variables may not have provided substantial new information, potentially due to redundancy or statistical dependence with SM20. Notably, Lorenz et al. (2021) used a simple analytical model framework to examine S2S forecasts of soil moisture. They showed that predictive skill largely originates from the persistence of land surface states (e.g., soil moisture anomalies), especially at longer lead time. Their analysis revealed that beyond 2 weeks, additional atmospheric inputs contributed marginally to skill, as autocorrelation in the land surface variables became the dominant source of predictability. This study aligns with our own findings as we have shown that additional predictors introduced minimal new information, and their relationships with soil moisture are often embedded in the SM20 signal itself. As such, the model may have learned to extract most of the relevant features for soil moisture forecasting directly from SM20. Thus, the remainder of this section will focus exclusively on the results achieved using SM20 as the input.

We further investigate the spatial variability of model performance by analyzing the bias and the ACC grid-wise (see Figure 3), as averaging over the entire domain can hide differences at individual grid cells and may not capture the spatial forecast variability over different lead times. As illustrated in Figure 3 (top), the ECMWF S2S model exhibits notable regional biases, particularly in Northern and Southern Europe, with positive and negative deviations that become more pronounced at longer lead times. In contrast, the DL models show considerably reduced biases, with near-zero deviations in most regions, proving higher skill and consistency across lead times. Overall, the results highlight the ability of the DL models to minimize biases and generalize effectively, particularly at longer lead times.

Figure 3 (bottom) illustrates the spatial variability of the ACC for the ECMWF S2S and the DL models. Consistent with the previous results, the skill remains generally higher for the DL, particularly in regions where the ECMWF S2S model struggles to maintain predictive accuracy. As expected, the ACC scores degrade with lead time, becoming increasingly evident beyond lead time 20 (corresponding to week 3, i.e., day 20). Additionally, the spatial differences in skill reveal regions of higher and lower forecast reliabilities. For instance, certain areas in central and Southern Europe maintain a higher skill, suggesting better model performance in these regions. In contrast, parts of Northern Europe and regions with complex terrain, such as the Alps, show more pronounced lower ACC scores with lead time. Despite the regional limitations, our results emphasize the potential of DL models to enhance forecast skill for soil moisture at extended time scales.

3.2. Seasonal Variability in the Forecasts Skill

We further investigate seasonal and regional variability in the skill scores by analyzing the bias and the ACC calculated for the each forecast initialization within each season: March–April–May, MAM; June–July–August, JJA; September–October–November, SON; and December–January–February; DJF.

The seasonal biases reveal similar trends in most seasons (see Figures S5–S8 in the Supplement), with the ECMWF S2S model showing higher negative bias, while the DL models display lower bias across most seasons. Moreover, we observe that the DL models tend to show higher and positive biases at longer lead times (e.g., week 3), particularly in central and eastern Europe, in MAM and JJA (Figures S6 and S7 in Supporting Information S1). This indicates a potential limitation in the ability of DL models to accurately represent the SM20 during the warmer months.

As for the bias, we further examined the seasonal ACC scores. The highest and positive correlations over most of Europe are shown in DJF (Figure 4, top), and the lowest correlations occur in MAM (Figure 4, bottom). For both seasons, a distinct regional pattern across lead times and models can be noticed. While in DJF, high values (>0.9) are found in central and southern Europe, the northern regions show lower ACC scores, being negative for the ECMWF S2S from lead time 20, and degrading for the DL models. At longer lead times 30 and 40 (corresponding to weeks 5 and 6), the skill decreases in central and southern regions (Figure 4, top). As illustrated in Figure 4 (bottom), the lower predictability in MAM is reflected by reduced ACC scores, which decline across lead times. Despite the improved forecasts produced by the DL models, the seasonal analysis suggests that forecasts initialized during spring tend to be less skillful. This seasonal dependence of the S2S forecasts skill has been the focused of previous studies (Dutra et al., 2021; Lesinger et al., 2024). For instance, Dutra et al. (2021) explored the skill of S2S forecasts in spring and summer of several surface variables, including soil moisture. While the

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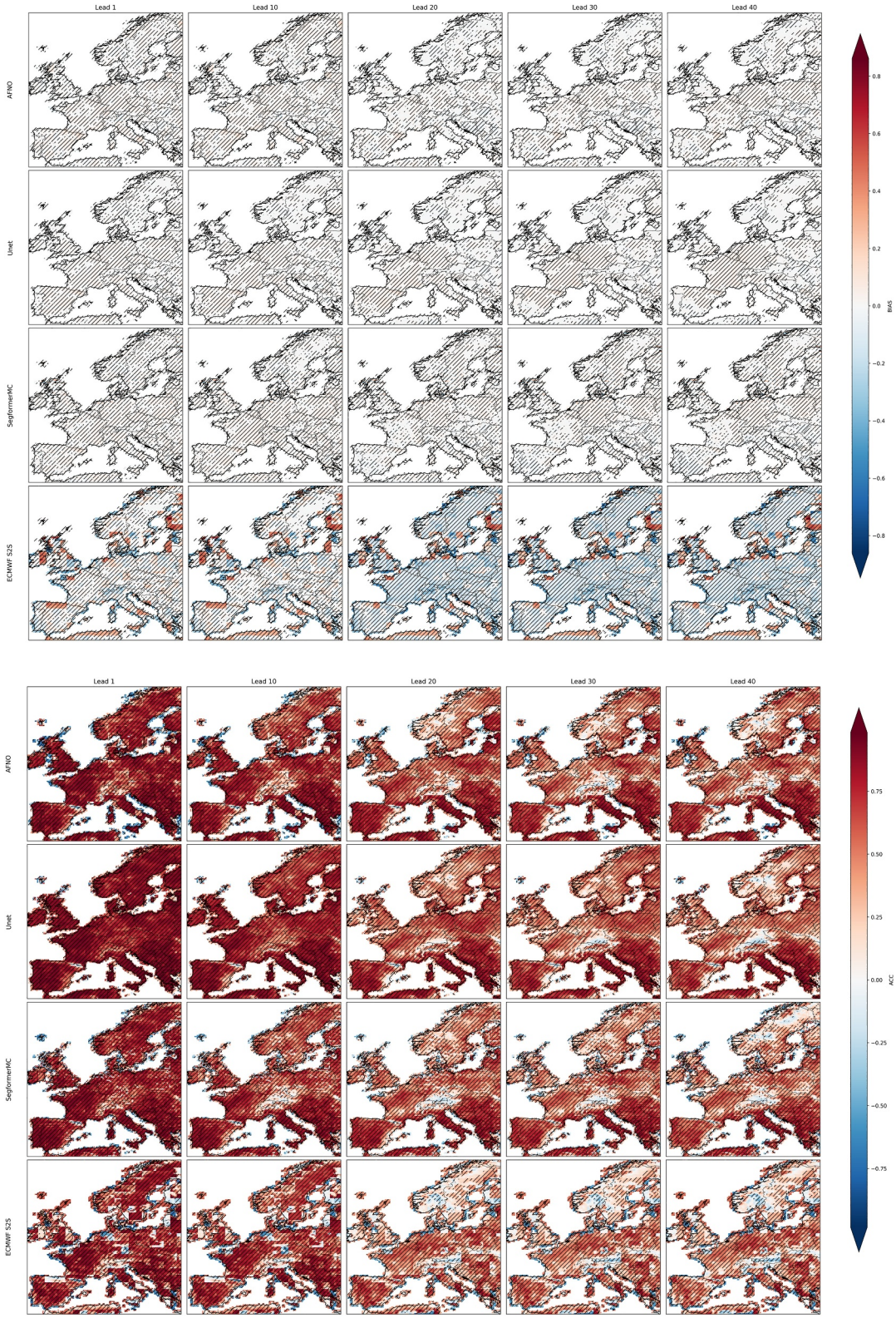


Figure 3. Comparison of spatial skill scores (bias at the top and the anomaly correlation coefficient at the bottom) across models and the following lead times: 1, 10, 20, 30, and 40. Hatched areas indicate where the skill score is statistically significant, calculated via bootstrapping with 10,000 samples at the 95% confidence level.

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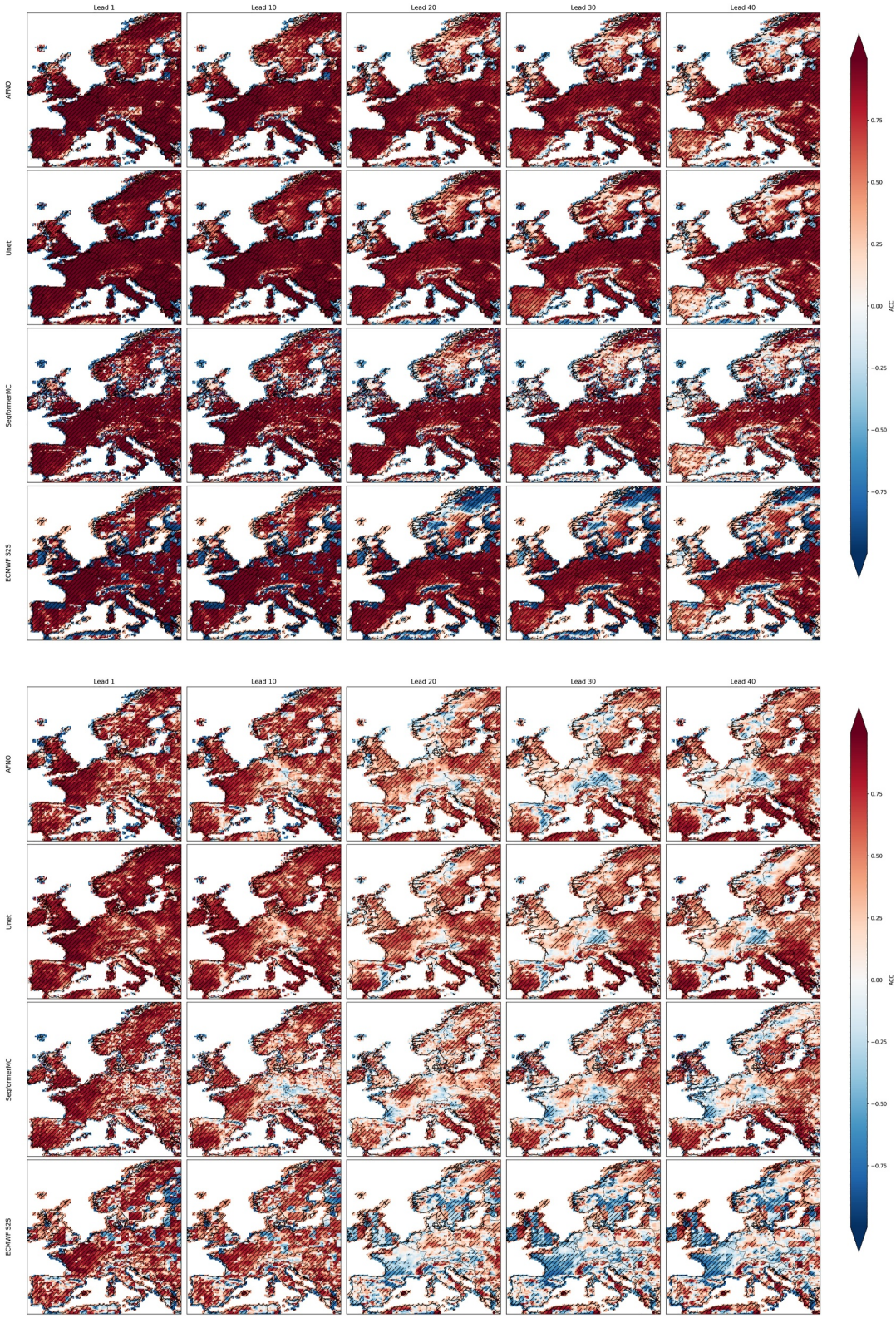


Figure 4. Spatial anomaly correlation coefficient compared across models and the following lead times: 1, 10, 20, 30, and 40, calculated for DFJ (top) and MAM (bottom). Hatched areas indicate where the skill score is statistically significant, calculated via bootstrapping with 10,000 samples at the 95% confidence level.

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authors did not find a robust seasonal dependence of the forecast skill, that is, they did not observe a strong association between regions with high bias and low forecast skill, they showed considerable deviations in soil moisture evolution with forecast lead times. Similarly, we observe decreasing skill with lead times and the regional differences across Europe when comparing seasons. In spite of a decreasing skill in MAM and JJA, our results show that the DL models outperformed the ECMWF S2S model even at longer lead times, which highlight the potential of neural architectures to correct the bias and enhanced the spatial resolution.

4. Conclusions

This work explores the potential of DL architectures for postprocessing subseasonal forecasts of soil moisture, a crucial variable with significant implications for regional flash drought early warning systems (Lesinger et al., 2024; Otkin et al., 2018). By addressing key shortcomings of numerical weather models—such as biases and coarse spatial resolution—our postprocessing framework demonstrates the value of advanced DL approaches to improve forecasts accuracy and to enhance their end-user usability. While DL models have become increasingly popular for applications, such as weather forecasting short-medium range and climate downscaling, their application to subseasonal forecasting remains limited (Horat & Lerch, 2024).

We conducted extensive experiments using three different DL architectures, including U-Net-based, SegFormer, and AFNO. The modeling framework is designed to be able to handle different lead times, which can be a daily or weekly time resolution. The models were trained on both daily and aggregated weekly data sets. Furthermore, we included additional predictors to assess their impact in the models performance. Our results revealed a similar performance when the DL models were trained on daily data compared to the trained models with weekly averages, suggesting that the temporal resolution did not significantly alter the overall predictive capability. Except for the SegFormer model, which exhibited a degradation in performance when using weekly aggregates, highlighting its sensitivity to temporal resolution, the other models showed similar performance. While we conducted several ablation studies on selected hyperparameters (e.g., dropout rates), a comprehensive hyperparameter optimization for each model was not performed. Therefore, we encourage additional benchmarks to better understand the impact of data aggregation on model performance,

Interestingly, our analysis demonstrated that including more predictors did not yield to a better performance, with only little marginal improvement. This suggests that the models may have already captured the critical information needed for accurate soil moisture forecast from the low-resolution input, or that the additional variables introduced redundancy or noise rather than contributing meaningful improvements. We also hypothesize that this behavior could partially be due to overfitting, given the relatively limited size of the data set, even when using data augmentation techniques. Similar observations were reported by Springenberg et al. (2025), who found that adding more inputs did not substantially enhance S2S wind speed forecasts using diffusion models. This highlights the need for a balanced approach when selecting predictors, ensuring that added complexity is justified by the availability of sufficient training samples, especially within the context of subseasonal forecasts.

While the DL models considered in this work demonstrated robust performance, some challenges remain. Seasonal variability in accuracy, particularly during spring and summer, indicates the need to refine model architectures and strategies to better capture the temporal and regional dynamics of soil moisture. Future work should focus on improving model generalization across seasons and exploring ensemble postprocessing approaches that incorporate higher-resolution observational data. Additionally, fine-tuning hyperparameters for individual architectures could help maximize the potential of each model and address observed sensitivities, such as those in the SegFormer model.

Finally, we would like to acknowledge that the model cycle updates could potentially influence downstream applications such as soil moisture correction. However, the bootstrapping analysis, which was used to statistically assess variability, confirmed the robustness of our results across different model cycles. As such, we believe this postprocessing framework can be applied directly to operational ECMWF extended-range forecasts without a significant loss in skill. Furthermore, retraining or fine-tuning the model maybe recommended following major updates to the forecasting system (e.g., substantial model cycle changes).

The work presented here can be readily adapted to other variables and higher-resolution data sets and can be transferred to other regions, offering a pathway toward ensemble postprocessing of S2S forecasts.

Data Availability Statement

The ERA5 data can be obtained from the Climate Data Store: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview.

Alternatively, it can be accessed directly at https://console.cloud.google.com/storage/browser/weatherbench2.

The S2S data are available at https://apps.ecmwf.int/datasets/data/s2s-realtime-instantaneous-accum-ecmf/levtype=sfc/type=cf/.

The code and datasets supporting this study are available at https://gitlab.hhi.fraunhofer.de/ai-aml/s2sdl.

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